

HALAC: Hybrid Adaptive Low-Latency Audio Codec

Research Project Roadmap

Project Overview

Title: HALAC: A Hybrid Adaptive Low-Latency Audio Codec with Machine Learning-Driven Content Awareness for Wireless Streaming

Duration: 6-8 months

Objective: To design, implement, and evaluate a novel audio codec that combines traditional psychoacoustic coding with lightweight machine learning for adaptive path selection, targeting high-fidelity wireless audio streaming applications.

Target Publication: ICASSP 2027 or IEEE/ACM Transactions on Audio, Speech, and Language Processing

1. Research Motivation and Gap Analysis

Current State of Audio Codecs

- **LDAC:** Achieves high quality at 990 kbps but with higher latency (approximately 10ms) and fixed algorithmic approach
- **Opus:** Optimized for VoIP scenarios with hybrid SILK+CELT architecture, but limited quality ceiling for high-fidelity applications
- **LC3:** Designed for Bluetooth LE Audio with low complexity, targeting lower fidelity
- **Neural Codecs (SoundStream, Encodect):** Target ultra-low bitrates (3-24 kbps) for speech applications, not suitable for high-fidelity streaming

Research Gap

No existing codec addresses the specific requirements of **high-fidelity wireless streaming** (Bluetooth, WiFi) with simultaneous optimization for perceptual transparency, low latency (sub-5ms), and adaptive bitrate efficiency. Current solutions make suboptimal trade-offs for emerging use cases including gaming audio, live streaming, and conversational media.

2. Proposed Solution and Key Innovations

Technical Approach

Hybrid architecture combining traditional MDCT-based psychoacoustic coding with lightweight machine learning-driven content classification and adaptive encoding path selection.

Target Specifications

- **Sample Rate:** 48 kHz stereo
- **Bitrate:** 400-800 kbps (variable bitrate, average 600 kbps)

- **Latency:** Less than 5ms total (encode + decode)
- **Quality Targets:** MUSHRA score > 80, PEAQ ODG > 0 (perceptually transparent)
- **Computational Complexity:** Less than 5% single CPU core utilization

Key Innovations

Innovation 1: Real-Time Content Classification

- Lightweight machine learning classifier (less than 50KB model size)
- Analyzes audio characteristics every 500ms
- Classifies content into categories: Speech, Music, Mixed, Transient-heavy
- Inference time budget: Less than 1ms per decision

Innovation 2: Adaptive Multi-Path Encoding Architecture

- **Path A (Speech-optimized):** Linear prediction + vector quantization, 3ms latency
- **Path B (Music-optimized):** MDCT + psychoacoustic model, 6ms latency
- **Path C (Hybrid):** Frequency-domain splitting with dynamic allocation, 5ms latency
- Real-time path selection based on ML classifier output

Innovation 3: Dynamic Perceptual Optimization

- ML-based prediction of Just Noticeable Difference (JND) thresholds
- Adaptive bit allocation targeting perceptual transparency
- Variable bitrate operation optimized for content complexity

3. Differentiation from Existing Work

Comparison with SoundStream (Google, 2021)

- **SoundStream:** End-to-end neural codec, 3-18 kbps, 24 kHz, targets VoIP applications
- **HALAC:** Hybrid traditional+ML, 400-800 kbps, 48 kHz, targets high-fidelity wireless streaming
- **Key Difference:** Different bitrate regimes (20-50x difference), different use cases, explicit latency guarantees

Comparison with LDAC

- **LDAC:** Fixed algorithm, 990 kbps, approximately 10ms latency
- **HALAC:** Adaptive ML-driven, 600 kbps average (40% reduction), less than 5ms latency
- **Key Difference:** Bitrate efficiency through intelligent adaptation, lower latency

Comparison with Opus

- **Opus:** Heuristic mode switching, VoIP-optimized
- **HALAC:** ML-driven path selection, high-fidelity streaming-optimized
- **Key Difference:** Finer-grained adaptive control, higher quality ceiling

4. Project Timeline and Milestones

Month 1-2: Research and Design Phase

Week 1-2: Literature Review and Prior Art Analysis

- Systematic search of IEEE Xplore, ACM Digital Library, arXiv
- Focus areas: High-fidelity wireless codecs, hybrid ML approaches, adaptive encoding
- Patent search: Qualcomm, Sony, Apple, Google, Fraunhofer
- **Deliverable:** Annotated bibliography with 50+ papers, overlap assessment spreadsheet

Week 3-4: Technical Specification and Architecture Design

- Detailed system architecture design
- ML classifier feature selection and model architecture
- Encoding path algorithm specification
- Bitstream format definition
- **Deliverable:** 10-15 page technical specification document

Week 5-6: Dataset Preparation and Tool Setup

- Collect diverse audio datasets (EBU SQAM, speech corpora, music samples)
- Manual content labeling for classifier training (500+ samples)
- Development environment setup: FFmpeg, TensorFlow Lite, CMake
- **Deliverable:** Labeled dataset, configured development environment

Week 7-8: Baseline Implementation

- Implement MDCT transform (encode/decode)
- Basic psychoacoustic model implementation
- Verify lossless reconstruction before quantization
- **Deliverable:** Working MDCT transform with unit tests

Month 3-5: Core Development Phase

Month 3, Week 1-2: Single-Path Encoder

- Implement Path B (music-optimized) as baseline
- Scalar quantization and Huffman entropy coding
- Bitstream packing and unpacking
- Initial testing on music samples
- **Deliverable:** Functional single-path codec at fixed bitrate

Month 3, Week 3-4: Decoder and Quality Verification

- Implement corresponding decoder
- Objective quality measurement (PEAQ, SNR)
- Bitrate measurement and analysis
- **Deliverable:** Complete encode/decode pipeline with quality metrics

Month 4, Week 1-2: Feature Extraction and ML Training

- Real-time audio feature extraction (MFCC, spectral centroid, zero-crossing rate)
- Convolutional neural network training (target: 85%+ accuracy)
- Model optimization and quantization
- Export to TensorFlow Lite format
- **Deliverable:** Trained ML classifier with validation results

Month 4, Week 3-4: Path A Implementation and Integration

- Implement speech-optimized encoding path
- Integrate ML classifier with encoder
- Path selection logic implementation
- Testing on mixed content
- **Deliverable:** Adaptive codec with two functional paths

Month 5, Week 1-2: Optimization and Path C Implementation

- Implement hybrid encoding path
- Bitrate optimization across all paths
- Quantization parameter tuning
- **Deliverable:** Complete three-path adaptive system

Month 5, Week 3-4: Performance Optimization

- Code profiling and bottleneck identification
- Latency optimization (target: less than 5ms)
- CPU usage optimization
- Memory footprint reduction
- Bug fixing and stability improvements
- **Deliverable:** Optimized prototype meeting target specifications

Month 6-7: Evaluation and Testing Phase

Month 6, Week 1-2: Objective Evaluation

- PEAQ testing on 100 diverse audio samples
- ViSQOL testing for speech content
- Latency measurements (encode time, decode time, total latency)
- Bitrate statistics (minimum, maximum, average, variance)
- CPU profiling and complexity analysis
- **Deliverable:** Comprehensive objective metrics dataset

Month 6, Week 3-4: Comparative Benchmarking

- Encode test samples with LDAC, Opus, LC3 at various bitrates
- Side-by-side objective comparisons
- Generate comparison tables and plots
- **Deliverable:** Comparative analysis report

Month 7, Week 1: Subjective Test Preparation

- Select 10 test stimuli (diverse content: speech, music genres, mixed)
- Encode with all codecs at multiple quality levels
- Implement MUSHRA test interface
- Prepare test administration protocol
- **Deliverable:** Complete test stimulus set and testing framework

Month 7, Week 2-3: Subjective Listening Tests

- Recruit 20+ test subjects (diverse demographics)
- Conduct MUSHRA listening tests in controlled environment
- Data collection and validation
- **Deliverable:** Raw subjective test data

Month 7, Week 4: Statistical Analysis

- ANOVA and post-hoc statistical tests
- MUSHRA score aggregation and confidence intervals

- Generate result visualizations (boxplots, scatter plots)
- ABX distinguishability analysis
- **Deliverable:** Complete subjective evaluation results

Month 8: Paper Writing and Submission

Week 1: Draft Manuscript

- **Section 1: Introduction (2 pages)**
 - Problem statement and motivation
 - Research gap analysis
 - Contributions summary
- **Section 2: Related Work (1.5 pages)**
 - Traditional codecs (LDAC, Opus, LC3)
 - Neural codecs (SoundStream, Encodec, Lyra)
 - Hybrid approaches and positioning
- **Section 3: Proposed Method (3 pages)**
 - System architecture
 - ML classifier design
 - Encoding path algorithms
 - Design trade-offs and justifications

Week 2: Results and Analysis

- **Section 4: Evaluation Methodology (1.5 pages)**
 - Objective metrics description
 - Subjective test protocol
 - Comparison methodology
- **Section 5: Results (2 pages)**
 - Objective results with tables and figures
 - Subjective test results
 - Comparative analysis
- **Section 6: Discussion (1 page)**
 - Strengths and limitations
 - Trade-off analysis
 - Use case recommendations
- **Section 7: Conclusion and Future Work (0.5 pages)**

Week 3: Manuscript Refinement

- Internal review and revision
- Figure and table refinement
- References compilation and formatting
- Abstract and title finalization
- **Deliverable:** Complete draft manuscript

Week 4: Final Preparation and Submission

- Address reviewer feedback (if internal review)
- Final proofreading
- Format for target venue (ICASSP/TASLP)
- Supplementary material preparation (code, data)
- Submission
- **Deliverable:** Submitted manuscript

5. Resource Requirements

Personnel

- **Primary researcher(s):** 1-2 persons
- **Advisor/supervisor:** 1 person
- **Listening test subjects:** 20+ volunteers

Computing Resources

- Development workstation (standard CPU, 16GB+ RAM)
- ML training: GPU recommended but not required (small model)

- Storage: 100GB+ for audio datasets and test results

Software and Tools

- **Development:** GCC/Clang, CMake, Git
- **Audio processing:** FFmpeg/libav, PortAudio
- **ML framework:** TensorFlow/TensorFlow Lite or PyTorch/ONNX Runtime
- **Testing:** PEAQ implementation, ViSQOL, custom MUSHRA framework
- **Analysis:** Python (NumPy, SciPy, Matplotlib), R for statistics

Data and Datasets

- EBU SQAM (Sound Quality Assessment Material)
- MUSHRA test signals
- Speech datasets (LibriSpeech, VCTK)
- Music datasets (diverse genres)
- **Total volume:** Approximately 50GB

Equipment

- Reference headphones for listening tests
- Quiet testing environment
- Audio playback equipment

6. Success Criteria and Expected Outcomes

Technical Success Criteria

- Working prototype meeting all target specifications
- **Latency:** Less than 5ms measured end-to-end
- **Quality:** MUSHRA score greater than 80
- **Quality:** PEAQ ODG greater than 0 (imperceptible difference)
- **Bitrate:** Average 600 kbps with perceptual transparency
- **CPU overhead:** Less than 5% single core

Academic Success Criteria

- Novel contribution clearly differentiated from existing work
- Rigorous evaluation with both objective and subjective metrics
- Acceptance at ICASSP or equivalent venue
- Open-source code release for reproducibility

Expected Impact

- **Academic:** Demonstrates feasibility of hybrid ML+traditional codec approaches
- **Practical:** Addresses gap in high-fidelity wireless streaming
- **Industry:** Potential adoption for gaming audio, VR/AR applications, podcast streaming

7. Risk Assessment and Mitigation

Risk 1: ML Classifier Accuracy Insufficient

- **Mitigation:** Start with proven features (MFCC, spectral), use established architectures
- **Fallback:** Reduce to binary classification (speech vs non-speech) if multi-class fails

Risk 2: Real-Time Performance Not Achieved

- **Mitigation:** Profile early and optimize incrementally
- **Fallback:** Accept slightly higher latency (7-8ms) if necessary

Risk 3: Quality Targets Not Met at Target Bitrate

- **Mitigation:** Tune psychoacoustic model carefully, consider higher bitrate ceiling
- **Fallback:** Adjust bitrate target to 700-750 kbps if 600 kbps insufficient

Risk 4: Subjective Test Logistics

- **Mitigation:** Plan early, use online crowdsourcing platforms if needed
- **Fallback:** Reduce number of subjects to 15 minimum (still statistically valid)

Risk 5: Prior Art Discovery

- **Mitigation:** Thorough literature search completed early (Month 1)
- **Fallback:** Pivot to comparative study or implementation focus if necessary

Risk 6: Timeline Overrun

- **Mitigation:** Monthly milestone reviews, scope reduction if needed
- **Fallback:** Drop Innovation 3 (perceptual targeting) to save 3-4 weeks

8. Deliverables Summary

Month 2

- Annotated bibliography (50+ papers)
- Technical specification document (10-15 pages)
- Labeled audio dataset (500+ samples)

Month 5

- Working prototype codec implementation
- Source code repository with documentation

Month 7

- Objective evaluation results (comprehensive metrics)
- Subjective test results (20+ subjects, MUSHRA methodology)

Month 8

- Research paper manuscript (6-8 pages, conference format)
- Supplementary materials (code, data, additional figures)
- Conference submission

Post-Submission

- Open-source code release (GitHub)
- Demo application (optional, if time permits)

9. Long-Term Vision and Future Work

Immediate Extensions (Post-Publication)

- Production optimization (ARM NEON, SIMD optimization)
- Bluetooth A2DP integration
- Mobile platform implementation (Android/iOS)
- Real-time streaming demo application

Research Extensions

- End-to-end optimization of ML classifier and codec jointly
- Multi-channel support (5.1, 7.1 surround sound)
- Packet loss resilience mechanisms
- Forward error correction integration

Industry Collaboration Potential

- Bluetooth SIG standardization consideration
- Gaming headset manufacturers (low-latency audio)
- VR/AR platforms (spatial audio streaming)
- Podcast platforms (speech+music optimization)

10. Publication Strategy

Primary Target: ICASSP 2027

- **Submission deadline:** October 2026 (estimated)
- **Notification:** January 2027
- **Conference:** May 2027
- **Rationale:** Premier venue for audio signal processing, strong ML integration

Backup Target: EUSIPCO 2027

- European conference, similar scope
- Slightly easier acceptance rate
- Good international visibility

Journal Target: IEEE/ACM TASLP

- Submit extended version post-conference
- More comprehensive evaluation and analysis
- Higher impact factor and citation potential

Dissemination Plan

- Conference presentation with live demo
- arXiv preprint simultaneous with submission
- Open-source release upon acceptance
- Blog post and technical documentation
- Social media announcement (academic channels)

11. Alternative Timeline: Accelerated 4-5 Month Plan for Late 2026 Publication

Overview

The standard 6-8 month timeline targets ICASSP 2027 (October 2026 submission). However, if publication in 2026 is required, an accelerated timeline is possible with strategic scope reduction and increased risk tolerance.

Target Venues for Late 2026

Primary Target: AES International Conference (November 2026)

- **Submission deadline:** July-August 2026
- **Notification:** September 2026
- **Conference:** November 2026
- **Rationale:** Audio engineering focused, accepts work-in-progress if rigorous

Backup Target: Regional Conferences

- IEEE Region 10 Conference (TENCON 2026)
- National Conference on Communications (NCC 2026)
- **Submission deadlines:** June-August 2026

Journal Option: Direct Submission

- IEEE Signal Processing Letters (fast review cycle: 2-3 months)
- AES Journal (4-6 month review)
- No fixed deadlines, rolling submission

Accelerated Timeline Breakdown

Month 1 (January 2026): Rapid Research Phase

- **Week 1:** Focused literature review (20-30 key papers only)
- **Week 2:** Technical specification (streamlined, 5-8 pages)
- **Week 3:** Dataset preparation and tool setup
- **Week 4:** Baseline MDCT implementation

Month 2 (February 2026): Core Development

- **Week 1-2:** Single-path encoder implementation (Path B only)
- **Week 3:** Decoder implementation
- **Week 4:** ML classifier development (simplified binary: speech vs music)

Month 3 (March 2026): Integration and Optimization

- **Week 1-2:** ML integration with encoder
- **Week 3:** Path A implementation (speech-optimized)
- **Week 4:** Performance optimization and bug fixing

Month 4 (April 2026): Evaluation

- **Week 1-2:** Objective testing (PEAQ, latency, bitrate)
- **Week 3:** Comparative benchmarking (vs LDAC, Opus)
- **Week 4:** Subjective testing (15 subjects, streamlined MUSHRA)

Month 5 (May-June 2026): Paper Writing

- **Week 1-2:** Draft manuscript (4-6 pages for conference)
- **Week 3:** Internal review and revision
- **Week 4:** Final preparation and submission (July deadline)

Scope Reductions Required

What Gets Cut:

1. Innovation 3 Dropped: Dynamic perceptual optimization removed entirely

- Focus only on content classification and dual-path encoding
- Reduces implementation time by 3-4 weeks

2. Path C Eliminated: Only implement Path A (speech) and Path B (music)

- No hybrid path implementation
- Binary classification sufficient (simpler ML model)

3. Reduced Testing:

- Subjective tests: 15 subjects instead of 20+
- Test stimuli: 6-8 samples instead of 10
- Fewer comparison codecs: LDAC and Opus only (skip LC3)

4. Simplified ML Model:

- Binary classifier (speech vs non-speech) instead of multi-class
- Simpler features (MFCC + ZCR only)
- Faster training, lower accuracy requirement (80% vs 85%)

5. Limited Optimization:

- Basic profiling only, no extensive optimization
- Accept 6-7ms latency if necessary (vs target 5ms)
- Focus on correctness over peak performance

Revised Target Specifications

Relaxed Targets:

- **Sample Rate:** 48 kHz stereo (unchanged)
- **Bitrate:** 500-800 kbps (higher average: ~650 kbps vs 600 kbps)
- **Latency:** Less than 7ms total (vs 5ms target)
- **Quality:** MUSHRA score > 75 (vs 80)
- **CPU overhead:** Less than 10% single core (vs 5%)

Pros and Cons Analysis

Advantages of Accelerated Timeline:

Pro 1: 2026 Publication Achieved

- Meets requirement for publication in current year
- Earlier graduation/milestone completion
- Faster CV building and career progression

Pro 2: Reduced Scope = Tighter Focus

- Binary classification easier to perfect
- Two-path system simpler to implement and debug
- Clearer narrative (less complex story)

Pro 3: Lower Resource Requirements

- Fewer test subjects needed (easier logistics)
- Less computing time for training
- Smaller dataset requirements

Pro 4: Faster Iteration

- Early feedback from reviewers
- Can extend work for journal version in 2027

- Learn from mistakes sooner

Disadvantages of Accelerated Timeline:

Con 1: Reduced Novelty

- Only 2 innovations instead of 3
- Binary classification less impressive than multi-class
- Reviewers may see as incremental rather than significant

Con 2: Higher Risk of Rejection

- Rushed evaluation may have gaps
- Smaller subjective study less statistically robust
- Less time to address unexpected issues

Con 3: Lower Quality Targets

- Relaxed specifications may not be competitive
- 75 MUSHRA score is "good" not "excellent"
- 650 kbps average closer to LDAC (less impressive savings)

Con 4: Limited Optimization

- Prototype may not demonstrate full potential
- Performance claims harder to defend
- May need significant rework for journal extension

Con 5: Compressed Evaluation

- 15 subjects borderline for statistical significance
- Fewer test stimuli = less comprehensive coverage
- Limited time for proper comparative analysis

Con 6: Stress and Burnout Risk

- Extremely tight deadlines
- Little room for error or setbacks
- May compromise quality under pressure

Risk Assessment for Accelerated Plan

Critical Risks:

Risk 1: Implementation Delays

- **Probability:** High (60-70%)
- **Impact:** Severe (miss submission deadline)
- **Mitigation:** Weekly milestone checkpoints, reduce scope further if needed

Risk 2: Quality Below Threshold

- **Probability:** Medium (40-50%)
- **Impact:** High (paper rejection)
- **Mitigation:** Focus on solid baseline, avoid overambitious claims

Risk 3: Insufficient Evaluation

- **Probability:** Medium (30-40%)
- **Impact:** Medium-High (weak results section)
- **Mitigation:** Prioritize subjective tests, use online crowdsourcing

Risk 4: Prior Art Discovered Late

- **Probability:** Low (10-20%)
- **Impact:** Severe (invalidates novelty)
- **Mitigation:** Complete literature review in Week 1 (non-negotiable)

Comparison: Standard vs Accelerated Timeline

Aspect	Standard (6-8 months)	Accelerated (4-5 months)
Submission	October 2026 (ICASSP 2027)	July 2026 (AES/Regional)
Publication	May 2027	November 2026
Innovations	3 (full proposal)	2 (reduced scope)
Encoding Paths	3 (Speech/Music/Hybrid)	2 (Speech/Music)
ML Classifier	Multi-class (4 categories)	Binary (2 categories)
Test Subjects	20+ subjects	15 subjects
Quality Target	MUSHRA > 80	MUSHRA > 75
Latency Target	< 5ms	< 7ms
Bitrate Average	600 kbps	650 kbps
Novelty Level	High	Medium-High
Risk Level	Medium	High
Success Probability	75-80%	50-60%
Venue Prestige	Very High (ICASSP)	Medium (AES/Regional)

Recommendation

For Maximum Quality and Impact: Follow the standard 6-8 month timeline targeting ICASSP 2027

Reasons:

1. ICASSP is the premier venue (3x citation rate vs regional conferences)
2. Higher success probability (75% vs 50%)
3. More complete implementation demonstrates full vision
4. Rigorous evaluation withstands reviewer scrutiny
5. Sets up strong foundation for journal extension

If 2026 Publication is Mandatory: Follow accelerated timeline but with clear understanding

Conditions:

1. Accept medium-tier venue (not top-tier ICASSP)
2. Accept higher rejection risk (50% chance)
3. Plan for journal extension in 2027 regardless
4. Commit to very strict weekly milestones
5. Have contingency plan if timeline slips

Hybrid Approach: Best of Both

Recommended Strategy:

1. Start with accelerated pace (January-May 2026)

- Aggressive development in first 4 months
- Get working prototype by May 2026

2. Assess progress in May 2026

- If ahead of schedule: Target AES November 2026
- If on schedule: Continue to ICASSP October 2026 submission
- If behind: Pivot to journal direct submission

3. Benefits:

- Maximum effort from start (no procrastination)
- Flexibility to adapt based on actual progress
- Multiple exit strategies depending on results

4. Concrete Decision Point:

- **End of Month 4 (April 2026):** Evaluate prototype quality
 - If prototype meets relaxed specs: Submit to AES (July deadline)
 - If prototype needs more work: Continue development for ICASSP (October deadline)

Conclusion on Timeline Options

Both timelines are viable but serve different goals. The standard 6-8 month plan maximizes quality and publication prestige, while the accelerated 4-5 month plan achieves 2026 publication at the cost of reduced scope and higher risk. The hybrid approach offers flexibility to adapt based on actual progress.

For a research project of this significance, rushing to meet an artificial 2026 deadline may compromise long-term impact. However, if external requirements (graduation, funding, career milestones) mandate 2026 publication, the accelerated timeline is achievable with disciplined execution and realistic scope management.

Final Recommendation: Pursue the hybrid approach - start aggressively but maintain flexibility to target the most appropriate venue based on actual progress and results quality.

clear differentiation from existing work, and commitment to open and reproducible research practices.

Project Team

- **Primary Researcher:** Joel
- **Secondary Researcher:** Adithya
- **Institution:** IIIT Kottayam
- **Advisor:** Dr. Ananth A

Document Version: 1.0

Date: January 7, 2026

Conclusion

This research project addresses a clear gap in high-fidelity wireless audio streaming through a novel hybrid approach combining traditional psychoacoustic coding with machine learning. The 6-8 month timeline is realistic with proper planning and execution. The project balances academic novelty with practical feasibility, targeting both publication and potential real-world impact. Success depends on rigorous evaluation,